

Title of the Project

AI-Powered Crop Disease Detection System

Problem Statement

Agriculture is a vital sector of the economy, but crop diseases can significantly reduce yield and quality if they are not identified at an early stage. Traditional disease detection methods rely on manual inspection by experts, which may not always be accessible or timely for farmers. An intelligent system that can analyze images of crop leaves and identify diseases using Artificial Intelligence (AI) can help farmers take preventive action quickly and improve agricultural productivity.

The proposed **AI-Powered Crop Disease Detection System** enables users to upload images of plant leaves. Using deep learning and computer vision techniques, the system detects diseases, identifies the affected crop, and provides basic recommendations for disease management.

Objectives of the Project

1. Develop an AI-based application to detect crop diseases from leaf images.
2. Automatically classify common crop diseases using deep learning models.
3. Provide users with disease information and basic preventive measures.
4. Create a simple and user-friendly interface for farmers and agricultural stakeholders.
5. Store disease detection records for future analysis and monitoring.
6. Promote early diagnosis to reduce crop losses and improve productivity.

Technology Stack

Programming Languages

- Python
- JavaScript

- HTML5
- CSS3

Frameworks

- React.js (Frontend)
- Flask or FastAPI (Backend API)
- TensorFlow/Keras (Deep Learning)

Tools

- OpenCV (Image Processing)
- Git & GitHub (Version Control)
- Postman (API Testing)
- Visual Studio Code (Development Environment)

Database

- MongoDB

Platforms & Services

- Docker (Optional for Containerization)
- Vercel (Frontend Deployment)
- Render or Railway (Backend Deployment)
- Kaggle Plant Disease Dataset or similar public datasets for model training

Domain Area of Application

- Agriculture
- Precision Farming
- AgriTech
- Artificial Intelligence
- Computer Vision

- Sustainable Farming

Expected Outcomes

- Accurate detection of crop diseases from leaf images.
- Faster diagnosis compared to manual inspection methods.
- Early identification of diseases, reducing crop damage and financial losses.
- Better decision-making through timely recommendations for disease management.
- Increased productivity and support for modern farming practices.
- A scalable AI solution that can be extended to support multiple crops and diseases.